

SentryAI: Video Anomaly Detection and Captioning

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Abstract. The rising prevalence of CCTV surveillance has yielded significant challenges in manual monitoring, with substantial costs, human error through fatigue, and delayed incident response. These issues highlight that it is imperative to develop automated systems in modern surveillance. This paper introduces, an exploration into how computer vision, deep learning, and natural language processing can automate video anomaly detection (VAD) through classification and captioning to enhance public security. Using the Multi-Scenario Anomaly Detection (MSAD) dataset, this study evaluates the performance of three models: CNNs with gradient boosting, Meta’s SlowFast architecture, and the transformer-based VideoMAE. Comparative analysis reveals that SlowFast achieves the strongest performance for both binary (F1: 65%) and multi-class (Macro F1: 21%) VAD, despite data imbalance. The SlowFast model is used to generate contextually and temporally relevant captions (achieving BLEU scores up to 0.30). The SlowFast architecture is recommended before discussing ethical considerations. Project code: [Github Repository](#).

Keywords: Computer vision, Deep Learning, Natural language processing, Video Captioning, Benchmarking models, AI for Public Safety, Anomaly Detection, Video Surveillance, Detection In Videos, Detection In Surveillance Videos, Real-world Anomaly, Real-world Anomaly Detection

1 Introduction

In the United Kingdom, there has been a long standing history with Closed-Circuit Television (CCTV) this technology has been implemented and utilised in wider society since 1960 by the Metropolitan police, this came as the service installed two temporary pan-tilt-zoom cameras in London’s Trafalgar Square, enabling the wide-scale monitoring of crowds during significant events such as the arrival of the Thai Royal Family and Guy Fawkes Day celebrations.

CCTV has seen significant developments since the 60’s, the use of the technology has expanded exponentially through both technological, and cultural developments. In 2013 a report by the British Security Industry Association (BSIA) estimated that there were between 4 million up to 5.9 million in the country, this equates to approximately one camera per every 11 to 14 people. Since 2013’s already significant expansion in growth, BSIA estimated in 2022 that this number

has grown to 21 million - suggesting there is now one CCTV camera for every three people. As evidenced through these figures, it comes as no surprise that the UK is one of the most surveilled nations globally.

The proliferation of CCTV technology yields unique challenges across the surveillance industry. Alternatives to manual analysis of extensive footage is becoming more imperative, as factors such as human fatigue, increased operational costs and delays in anomaly detection or incidents become more prevalent. These limitations prove progressively more detrimental in responding to critical incidents potentially posing risks to public safety. Through the implementation of automated services, surveillance efficiency can be enhanced monumentally, ensuring time-critical responses to potential risks.

This project incorporates computer vision and natural language processing to alleviate the complex issues arising from manual monitoring of large scale surveillance systems. Through exploration of CCTV data for preprocessing and applying deep learning architectures, this study provide recommendations on a preferred methodology that may be applied on a large scale to drastically improve time incident response, post-event analysis, and improved situational awareness for public safety.

To develop a robust anomaly detection and event description framework this project utilises the Multi-Scenario Anomaly Detection (MSAD) dataset [1] offers a rich variety of CCTV footage, encompassing a broad range of scenarios and camera viewpoints, by extension, this dataset is imperative for ensuring a trained model can reach a greater generalisation potential.

This study seeks to deliver a deployable deep learning model for anomaly detection and event captioning in CCTV footage, achieving this aim will yield superior real-time incident response and post event analysis. Achieving the task of building a robust model capable of real-time responses, the following objectives must be achieved:

- Investigate the performative potential of various computer vision models, this will include static feature extraction, end to end CNN models, along with transformer based models.
- Explore the application of NLP models to generate textual descriptions of the MSAD dataset which match the relevant context of the footage.
- Explore and evaluate the Multi-Scenario Anomaly Detection (MSAD) dataset, through exploratory data analysis and its application for real-world CCTV applications.
- Analyse and compare performance metrics of tested model utilising standard data science metrics of F1 score, Precision, Recall for classification of anomalies, BLEU for captioning.
- Discuss and explore the legal and ethical implications of AI-driven surveillance, covering the topics of privacy, data bias and security concerns.

2 Literature Review

Through the integration of deep learning and natural language processing into surveillance, the quality of advanced anomaly detection and contextual event understanding in CCTV footage has dramatically improved. This literature review will explore the fundamental studies and methodologies in Video Anomaly Detection (VAD), video captioning alongside cross-modal learning to achieve the objectives placed by the study. Starting with VAD, this methodology incorporates identifying anomalous patterns in surveillance footage that would correlate to an incident hazardous to wellbeing in surveillance footage, this will often utilise deep learning models due to their flexibility when applied to complex data. Traditional methods relied heavily of manually engineered features and statistical models which often lack generalising capabilities when applied across different environments. More recent approaches have attempted to apply deep learning architectures such as CNNs, RNNs and transformer-based models to optimise test performance.

The 2024 paper by Zhu et al. yields a key contribution to this field, this piece of work presents the Multi-Scenario Anomaly Detection (MSAD) dataset and evaluates multiple machine learning models for anomaly detection. The study highlights the challenges of generalising anomaly detection due to the diverse range of possible CCTV placements, the study was successful in safeguarding against overfitting by consolidating a large plethora of CCTV footage data over multiple scenarios [1]. Multiple viewpoints and scene complexity is imperative in this study as the end result seeks to be applied across the larger public domain.

A methodology to be explored through the study is the Feichtenhofer et al. (2019) SlowFast network implemented by Meta, this deep learning architecture processes video data utilising two separate pathways: a slow pathway which captures spatial semantics and a fast pathway which considers motion dynamics. Through the dual-pathway approach, models can yield state-of-the-art performance when applied to video classification and anomaly detection tasks [5]. Other methodologies related to this domain include autoencoders and generative adversarial networks (GANs), these learn latent representations of normal behaviours before flagging deviations as anomalies. This approach can be found in works such as Sabokrou et al. (2018) who demonstrated a reduction in false positive rates in surveillance video analysis [6]. GAN-based methods are often highly computationally expensive with extensive training for well-generalised model, this raises a level of unsuitability for deployment in a live emergency scenario. For the development of textual descriptions of the CCTV footage, this is a relatively recent area, pushed by advancements in vision-language models. Traditional methodology relied on rule based captioning which often failed to generalise across changing video contexts. Recent approaches utilise deep learning architectures ranging from Long Short-Term Memory (LSTM) networks, transformers, and cross-modal attention mechanisms.

Studies have investigated the integration of visual transformers with pre-trained language models for automated video captioning. Self-attention mechanisms in transfer architectures have been shown to boost contextual understand-

ing of images and video frames, enabling greater efficiency at producing coherent and relevant captions. Research in this field has demonstrated the effectiveness of merging spatiotemporal features with natural language models to enhance captioning quality. A major advancement in this field originated from OpenAI, the company achieved this through the launch of their CLIP (Contrastive Language-Image Pretraining) model which was adapted for video captioning tasks. CLIP’s zero-shot learning capabilities allow it to produce relevant textual descriptions while by-passing task-specific labeled data, this makes it a promising candidate for CCTV-based video captioning where labelled and annotated data may be scarce [8].

Combining the previous approaches, of anomaly detection and that of video captioning would require the use of cross-modal learning techniques, bridging the visual and textual modalities. One approach is the fusion of vision and text embeddings, this enables a model to associate detected anomalies with the relevant descriptive captions. A recent example of this is the VideoBERT, this jointly models video frames and textual descriptions to provide a contextual understanding of the video [9]. The TEVAD (Text Empowered Video Anomaly Detection) framework, this paper was introduced in 2023 by Chen et al. The paper focuses on automatic anomaly detection in surveillance systems, the novel framework utilises visual and textual features, the text attributes complement the visual features to create a semantic understanding of the video context, achieving SOTA performance [10].

Although significant advancements in VAD and video captioning have been made in recent years, through extensive literature review, it is apparent that several challenges yet remain. A major issue is the high volume of high false positive rates in anomaly detection, this will consequently lead to unnecessary alerts in real-world surveillance applications, ironically expending more time and money than would have been required without the system. Addressing this challenge through enhanced feature extraction techniques and incorporation of temporal dependencies is a strong possibility. A significant limitation in classification algorithms is the potential bias in datasets, this can result in performance and ethical issues if the training data does not represent the diversity of a real-world surveillance scenario, due to reduced model generalisational abilities. Datasets such as MSAD [1] has attempted to address this, however future research could expand on this further through synthetic data augmentation techniques to improve robustness. The ethical concerns additionally play a crucial role in AI-driven surveillance, privacy, bias and potential misuse of the technology. These factors are considered in further depth within this study to ensure responsible use of automated CCTV systems.

3 Methodology and Experiments

3.1 Exploratory Data Analysis

Dataset overview. The MSAD dataset is the primary knowledge base utilised within the study. This dataset has been sourced and permitted by the Australian

National University [1]. The data consists of 720 individual videos, accounting for a total frame count of 447,236 frames, a majority of videos contain a framerate of 30fps though some videos reach a low of 6fps. The dataset holds a large variety of video resolutions, ranging from 480x360 up to 1920x1080. The dataset is partitioned by anomaly and scenario with 12 (11 excluding the normality class) and 14 unique values respectively.

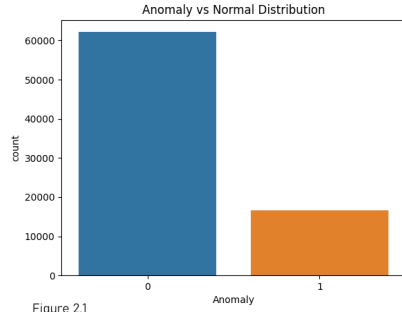


Fig. 1. Anomaly vs Non-Anomaly

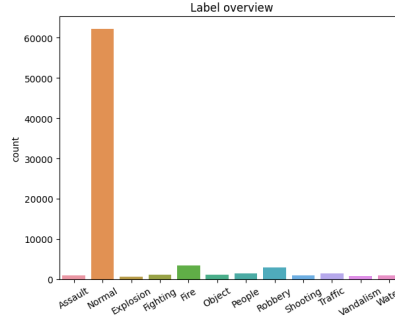


Fig. 2. Multi-scenario

Fig. 3. Class balances by binary and multiple scenario.

The scenarios metadata attribute covers locations, ranging from stationary views such as warehouses and front-doors, to more active environments such as roads, highways or trains. Anomalies incorporate a variety of situations which are considered hazardous to public safety, categories for fires, explosions and crashes are to name a-few. To preserve temporal information whilst reducing file size, the project extracts every 5th frame before applying normalisation transforms.

As illustrated in figures ?? and ?? there is a class imbalance, approximately 62,000 frames are attributed as normal compared to 18,000 frames which are anomalous, suggesting a near 3:1 ratio towards normal frames. The class imbalance exemplifies when partitioned by anomaly type, this is imperative as a highly imbalanced dataset needs to be accounted for in preprocessing and model training. Imbalance occurs further when MSAD is stratified by scenario, through an analysis of video frames, there is an absence of frames in the highway scenario suggesting that models may experience difficulty in generalisation on data in this scenario. Due to the variety in scenarios, there is sufficient contrast in camera perspective, this by extension yields greater variability to the training data, discouraging models from overfitting. Breakdown of scenarios partitioned by anomalies can be seen in figures 4 and 5.

To produce further insight into the separability of the data, we can reduce high dimensional data into formats we can visualise using manifold learning, this study has applied t-SNE for a reduction into lower dimensional space (2D) [2]. As displayed in the figure, anomaly types have clustered together which suggests

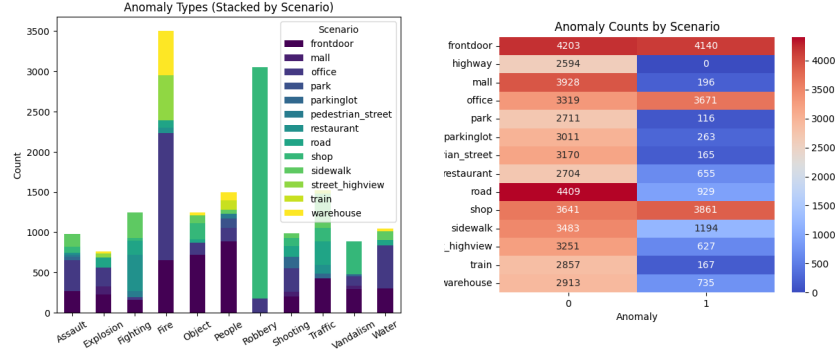


Fig. 4. Scenarios partitioned by multi-anomaly type.
Fig. 5. Scenarios partitioned by binary scenario anomaly type.

Fig. 6. Scenarios partitioned by anomalies

that models will theoretically be able to differentiate anomalies from normality. Challenges arise as clustering may have captured the temporal elements of each CCTV clip, further challenges lie in intersecting clusters where anomaly type is different.



Fig. 7. t-SNE by Anomaly Type

In summary (figure 7), the findings of the t-SNE visualisation technique yielded results that suggest that while some anomaly types form distinct clusters, others exhibit some degree of overlap, this suggests that there may be some performance issues on classification model training. Class imbalances may be

rectified through weighted loss functions, oversampling under represented classes or through synthetic data augmentation. The variability in camera perspectives and scene offers further complexity, enabling models to perform to greater accuracies upon deployment. These insights inform the choice of deep learning architectures and preprocessing methods that will ultimately enhance the final anomaly detection model in real-world CCTV applications..

3.2 Preprocessing

To achieve optimal results, it is imperative to preprocess data appropriately before model training. To efficiently process the dataset we first extracted every fifth frame to reduce redundancy while maintaining temporal structure of the footage. This helps in reducing computational costs without a significant impact on the quality of temporal feature extraction. To enable the use of 3D convolutional networks (CNN's), frames were concatenated into clips of 16 consecutive frames using a sliding window approach. This ensures that the model can capture temporal dependencies in short video segments making it suitable for anomaly detection in CCTV detection.

Upon extraction, a transform is applied to the frame, this allows standardisation for more effective capturing of features. Each frame is resized to 224x224, this enables faster training due to smaller computational memory requirements alongside aligning all dataset frame resolutions before passing through any given model. Frames were permuted from BGR format to RGB as this colour space is more frequently used in models.

In order to improve model generalisation and stability, it was imperative to implement normalisation to the transformed dataset. Pixel values were rescaled using mean = [0.485, 0.456, 0.406] and standard deviation = [0.229, 0.224, 0.225], this is akin to industrial standard values that was first implemented on the ImageNet dataset [3]. This normalisation ensures that the input distribution matches the pretrained weights of implemented models such as 3D CNN's, reducing issues such as varying lighting conditions or colour intensities from variation in CCTV clips. A further benefit in normalizing inputs is the reduced training biases and accelerated model convergence.

In addition to preprocessing frames, metadata was generated to track extracted clips and their corresponding features. Each video clip was assigned a unique index and labeled with its source video filename, ensuring that features could be easily linked back to their original footage. A structured CSV file was created to store this metadata, containing details such as video filenames, clip indices, and feature file paths. Furthermore, anomaly types were extracted from the video titles, allowing for better organisation and potential supervised learning applications in anomaly detection tasks.

From our EDA it was uncovered that there was significant imbalance once anomalies were partitioned by type. To account for this imbalance in classes, oversampling techniques have been applied. The fundamental objective of this paper is to detect anomalies from CCTV data, this would face significant challenges without accounting for the minority classes.

3.3 Method 1: Static CNN Extraction with Traditional Classification

The initial technique applied in the study was a static CNN feature extraction (Resnet) followed by traditional classification using an Extreme Gradient Boosting Classifier (XGBC). Through this methodology we establish a baseline performance with a computationally efficient yet effective method for anomaly detection in CCTV footage. Unlike future methodologies, this technique is not an end-to-end deep learning model which processes raw video frames, static feature extraction focuses on computational efficiency by extracting spatial representations using a pre-trained model before forwarding extracted attributes to a gradient boosting classifier for anomaly detection.

For the implementation, this involved extracting deep feature representation from video clips via the CNN-based model. The features are then fed into the fine-tuned XGBC, this methodology is optimised for handling imbalanced datasets due to the traditional ability of adjusting decision boundaries based on an ensemble of weak learners. This approach provides greater interpretability, reduced training times and enhanced generalisational abilities [4]. Through CNN feature extraction and GBC for classification this method ensures that time-sensitivity can be addressed through computational efficiency.

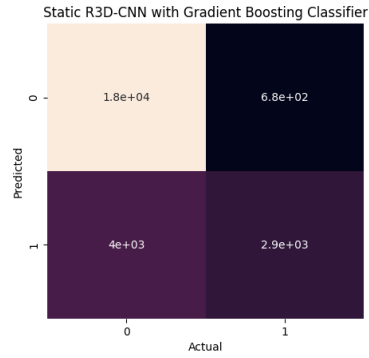


Fig. 8. XGBC Binary

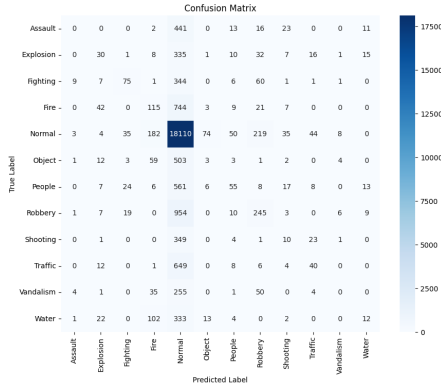


Fig. 9. XGBC Multi-Scenario

Fig. 10. Confusion Matrices

Upon completion of the binary model training, the XGB achieved an accuracy score of 82%, as can be seen in figure 8 while this value appears high, it is paramount to understand that due to the significantly imbalanced testing set, this result does not fully represent the “true” performance. The static CNN extraction with the application of a XGBC we have achieved an F1 score of 55% consisting of 42%, 81% from precision and recall respectively. The results suggest that this methodology yields moderate performance for the task of anomaly

detection. The high recall with lower precision implies that the model correctly detects most of the actual anomalies which is vital to the safety critical application, at the cost of a large number of normal cases becoming misclassified and potentially wasting human resources through unnecessary alerts.

From figure 9 we may infer that when this methodology is expanded to a multi-class scenario, the models’ performance suffers significantly, the model is excellent at filtering out normal videos, which is an imperative first step when building a VAD. The model yields extremely low recall values for nearly all anomaly classes which is dangerous in the context of security and heavily biased towards the “Normal” class despite the deployment of oversampling techniques, a major benefit of this methodology however is the speed, extreme gradient boosting algorithms are relatively much faster than deep learning models which is advantageous in real-time scenarios.

3.4 Method 2: SlowFast Network for Video Recognition

The SlowFast Network, originally presented by meta in 2019, is a state of the art deep learning architecture that has been designed for video understanding tasks, with a particular focus on action recognition and anomaly detection. This methodology has often been adopted due to its ability to efficiently process spatial and temporal features. The model receives two parallel channels: the slow pathway, this captures high-resolution spatial features at a low frame rate, on the contrary, the Fast pathway processes a high frame rate but with reduced spatial resolution. The parallel design enables the model to learn from a synergy of long-term context and short-term motions, yielding a suitable methodology for anomaly detection and visual context understanding.

When trained for the binary classification of anomalous events, the SlowFast Network has yielded promising results, from figure 11 with an overall accuracy score of 85% and an F1 score of 65%, the model with further finetuning would be suitable for deployment on a wider scale. Recall and precision metrics were at 62% and 68% respectively. While similar scores, the model holds a higher precision which suggests there is a level of conservatism with prediction anomalies, enabling fewer false alarms at the cost of potentially missing actual positives. Figure 12) informs us that when applying the slow-fast architecture to multi-scenario cases the model achieves an accuracy and weighted F1 score of 79% and 74% respectively. this is primarily caused through a significant imbalance in the dataset. The detection of normal events achieves an exceptionally high f1 score of 90% but this performance drops to an average macro f1 score of 21% once minority classes are introduced, this comprised of the macro averages of recall 19% and precision 35%. With an expansion of the MSAD dataset many of the under performing classes will see significant improvement in performance as additional data is supplied. Nonetheless, in the case of “random guessing”, we would expect a f1 score of 8% or even lower due to the significant imbalance of the dataset, this implicates that this model is successfully learning to differentiate between classes but leaves significant room for further development.

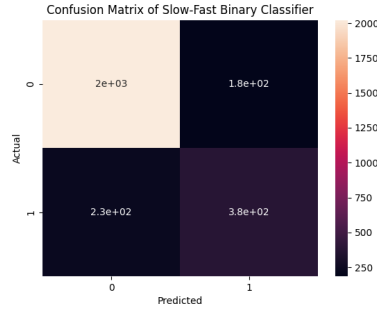


Fig. 11. SlowFast Binary



Fig. 12. SlowFast Multi-Scenario

Fig. 13. Confusion Matrices

3.5 Method 3: Transformer-Based Architecture VideoMAE

VideoMAE (Video Masked AutoEncoder) is model with transformer-based architecture, the methodology is designed for the learning of both spatial and temporal representations of video data through self-supervised training. The model is based on masked autoencoding strategies that are utilised in images and extended to the video domain through random masking of high proportions of video patches and training the model to reconstruct the missing content. VideoMAE divides input video clips into 3D patches (encompassing of time, height and width), encodes only the visible patches using a video transformer, this is then decoded for the prediction of the original appearance of the masked patches. This forces the model to learn meaningful representations of motion, video appearance alongside temporal context, this has proven to result in strong performance across downstream tasks such as video classification and anomaly detection [7].

This methodology holds relevance to the study as VideoMAE is capable of achieving good performance in VAD tasks. The ability to learn spatial-temporal features without extensive labelled data is well suited for the MSAD dataset, where normal clips vastly outnumber the minority classes which represent video anomalies, as represented through the exploratory data analysis. Through training the model to understand normal and anomalous visual patterns the model will be able to detect deviations associated with emergency response requiring behaviour.

When applied to the binary case, VideoMAE achieved poor results, and when training on the data, it overfitted easily (See Appenidx A 2.) The model yielded a moderate accuracy of 60% though the F1 score was significantly low, yielding only 36(Figure 14)

While performing marginally better then the binary case, the multiple scenario classifier also yielded subpar performance, though it showed potential for

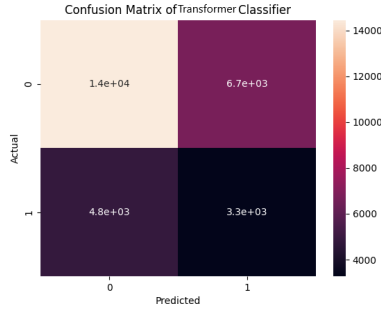


Fig. 14. VideoMAE Binary

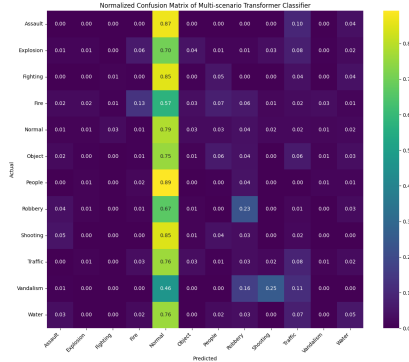


Fig. 15. VideoMAE Multi-Scenario

Fig. 16. Confusion Matrices

superior results with additional resources (See Appendix A 2.). The best performing classes were ‘Robbery’, ‘Fire’ and ‘Normal’ where the F1 scores reached 30%, 22% and 16% respectively, while these results are low, with further training and data the multiple scenario could excel in tasks specific to these performances. A primary example of deployment would be a warehouse or store, a fire or burglary could occur outside of operational hours, deploying the VideoMAE architecture would enable constant surveillance and quicker response times than a potentially unmanned facility. (Figure 15)

4 Conclusion and Recommendation

Table 1. Binary Classification Performance Table

Model	Accuracy	Precision	Recall	F1 Score
CNN + XGBC	82%	42%	81%	55%
SlowFast r50	85%	68%	62%	65%
VideoMAE	60%	33%	41%	37%

The comparative results for the binary case indicate varying performance amongst the classification models. Table 1 informs us that SlowFast r50 yielded the strongest overall performance with the highest overall accuracy (85%) and F1 score (65%), this indicates that the model offers a great balance between correctly identifying positive instances correctly and the minimisation of false positives. In contrast static CNN with XGBC achieves a high accuracy (82%) alongside the highest recall (81%), indicating it’s excellent at capturing positive incidences

at the cost of a lower precision (42%), meaning the model yields a high quantity of false positives. The VideoMAE model offered the weakest performance of the considered models with an accuracy of (60%), precision (33%), recall (41%) and F1 Score (37%), these metrics inform that the model struggles with not only positive cases but also avoiding false alarms.

When deploying an optimal model for binary VAD, the specific application and relative importance of precision and recall would dictate model choice. The slowfast architecture appears to be the most robust, while CNN + XGBC would be suitable if high recall is prioritised over false alarms. The poorer results of the VideoMAE suggests that this model would be less suited to this task without further optimisation.

Table 2. Multi-Scenario Classification Performance Table

Model	Accuracy	Precision	F1 Score
CNN + XGBC	86%	25%	14%
SlowFast r50	79%	35%	19%
VideoMAE	43%	15%	11%

The multi-case scenario highlights the issue of data imbalance as explored in the EDA section of the paper, with significant differences in F1 score when compared to their binary counterparts. From Table 2 we can see that overall, while the accuracy is moderately high, the class-level performances are considerably low. The CNN+ XGBC methodology achieves the highest accuracy at 86%, but the macro-averaged f1 score sits at 17% with recall (14%) and precision (25%), the direct cause of these figures is the bias towards classifying clips as normal (majority class), leading the model to perform poorly on minority classes. This suggests that whilst it classifies a majority of clips correctly overall, there are issues when generalising across all scenarios in a balanced way.

The SlowFast R50 Model held a slightly lower accuracy rate overall at 79%, while this metric was inferior, macro f1 score, precision and recall stood at 21%, 35% and 19% respectively, which were the highest metrics out of all models. This yields us a more balanced result than the CNN + XGBC model. The VideoMAE algorithm again underperformed, with an overall accuracy of just 43%, precision (15%), recall (11%) and f1 score (11%) were the worst performing metrics in the experiment though hold the potential for the greatest improvement if given an enhanced dataset.

Given that SlowFast R50 yielded the greatest results across all metrics, barring overall accuracy, it is the best choice for VAD. Through implementing further balancing techniques or architectural adjustments, this model has the potential to achieve significantly higher f1 scores. Due to the greatest performance across both classification tasks, SlowFast R50 stands out as the recommended methodology for the task of anomaly detection in CCTV footage.

4.1 SentryNet (Captioning System)

The Slow-Fast architecture proved superior to other considered models for the tasks of this study; for this reason, it was selected to be the backbone model of the CCTV Captioning system. The model has been designed with an encoder-decoder design, where features are extracted from the clips before being fed into a custom LSTM network with implemented attention in order to improve contextual understanding of CCTV footage. The model utilises 3 layers in order to capture the level of relevancy needed for a video captioning system whilst attempting to safeguard against overfitting by avoiding unnecessary complexity. With the Slow-Fast architecture Upon completion of training, the average BLEU

Table 3. Comparison of Generated vs. Original Captions with BLEU Scores

Generated Caption	Original Caption	BLEU Score
the the short kindergarten teacher wearing a red top and green apron and a little girl s hand,	the short kindergarten teacher wearing a red top and green apron covered the little girl s mouth with one hand and held down the little girl s body with the other hand to prevent her from moving the little girl s legs kept struggling	0.2810
the the woman in the black and white skirt picked up a pair of pants and the them on the shelf and the the pants, the the woman in the black and white skirt picked up a pair of pants and the them on the shelf and the man in the black and white skirt	the woman in the black and white skirt picked up a pair of pants and put it on her arm and took the pants from the bottom of the shelf to the top for counting	0.2504
the woman in the black and white skirt picked up a pair of pants and the them on the shelf and the man in the black and white skirt also walked to the the pants	the woman in the black and white skirt also picked up a pair of pants and looked at them and the woman in the red skirt put down her skirt jeosŭ	0.2320

score lies at 0.03. While this is a low average, the model sucessfully proved proof of concept that our Sentrynet architecture is capable of generating semantically meaningful and temporally grounded descriptions of real-life events in CCTV footage as can be seen in table 3. Captioning videos adds an additional complexity to computer vision models, SentryNet has sucessfully learned and captioned objects, in challenging scenarios which include rapid motion and multi-person interactions,, the model can correctly identify key characters with limited training data ("Kindergarten teacher", "Woman in black and white skirt"), alongside

actions and specific locations within the clip. This is imperative for security applications as this brings an additional level of context that the models can provide to the user. SentryNet has achieved a strong proof-of-concept that automated CCTV systems can successfully translate video content into natural language summaries.

5 Ethical Considerations

On the implementation of artificial intelligence into CCTV surveillance, it is imperative to consider the ethical repercussions that may arise and how to address responsible deployment. AI offers an automatic and cost effective methodology for anomaly detection and video captioning in surveillance applications, it also raises issues related to privacy, bias, accountability and transparency. This section of the study investigates fundamental ethical concerns in automated CCTV systems and mitigation strategies for risk minimisation.

A primary ethical concern in the deployment of AI-based CCTV is privacy infringement. Through machine learning, models analyse vast amounts of video footage, this will capture and process personal data, often without explicit consent. This raises questions about the right to privacy and potential misuse and function creep of surveillance data. The General Data Protection Regulation (GDPR) in Europe imposes strict guidelines and regulation on data collection, storage and processing. To comply with this framework, AI CCTV systems should incorporate privacy-preserving techniques, such as data anonymisation, encryption and differential privacy. For the purposes of this study, the MSAD dataset is utilised which incorporates facial blurring to ensure the privacy of any individual within the training data [1].

Bias in AI is a fundamental concern in any classification model. Within the domain of ethics, bias can lead to unintentional discriminatory surveillance practices, anomaly detection and facial recognition may exhibit biases based on race, gender, or socioeconomic status, this is often due to skewed and imbalanced training data. Studies have demonstrated that facial recognition technologies have higher error rates for individuals from minority groups due to proportionally less data, this can lead to misidentifications, false positives in detections and potential civil rights violations.

To mitigate bias, it is imperative to ensure diverse and representative data within the training datasets, other methodologies such as fairness aware algorithms could be incorporated with weights or oversampling applied to loss functions and datasets. Through the implementation of human oversight, this could additionally assist in reduction of the risk in automated discrimination before model deployment

Anomaly detection models often are met with the issue of false positives, flagging normal activities incorrectly as suspicious. This can result in unnecessary emergency service involvement, public distress and wrongful accusations. An example could be a model trained on limited behavioural patterns might

misinterpret normal activities as potentially endangering, leading to wrongful scrutiny of an individual or community.

An approach to account for this challenge could be through refining context understanding in the model, enabling environmental and temporal factors to be utilised to improve accuracy and reduce false positives. Another approach would be to have all positives filtered through human intervention, offering an efficient medium between fully automated and manual surveillance. Through widespread deployment of AI surveillance, it raises concerns regarding mass surveillance for authoritarian misuse. Governments and corporations may exploit mass surveillance for excessive monitoring through function creep, this can result with infringement on civil liberties and GDPR violations.

It is imperative to balance the aforementioned ethical concerns with the security-enhancing objectives of this study, therefore it is vital that systems are deployed with clear and ethical guidelines, independent reviews, and public consultations, assisting to prevent against overreach. The 2023 EU's AI act or IEEE's Ethical Considerations of Autonomous and Intelligent Systems is an excellent example of framework that can ensure responsible use of autonomous surveillance technology.

Deep learning models often operate as black boxes, which leaves difficulty in understanding of what leads the model to flag certain behaviours as anomalous. Lack of explainability can erode public trust and make it challenging to decipher reasonings behind a model's classification. Incorporating explainable AI techniques such as Grad-CAM may be integrated into deep learning visualisation models to improve interpretability and justifications for detections.

It is through these ethical considerations surrounding AI-based CCTV surveillance that models can be deployed ethically and responsibly. While AI offers significant advancements in security applications, deployment must be guided by strong ethical frameworks to prevent misuse and protect civil liberties.

6 Future Work

Future work will focus on boosting the capabilities of SentryNet for both classification (VAD) and captioning.

Firstly, the primary aim will be to explore a multi-modal approach through the integration of the video captioning model and the selected classification model. The integration has potential to provide a symbiotic performance boost through a more comprehensive understanding of anomalous events, with both visual elements and textual attributes, yielding a more comprehensive model, reducing false positives and boosting classification F1 Scores.

Secondly, by enhancing and expanding the MSAD dataset with other similar projects alongside the creation of new data, we can create a substantially larger and more diverse set of data. Additional research into synthetic data augmentation could further yield variety in the dataset. The result of this task should provide the model a greater ability to generalise across a wider range of scenarios.

Finally future work will incorporate explainable AI (XAI) techniques such as Grad-CAM (Gradient-weighted Class Activation Mapping), this will not only provide greater insight into the models decision-making processes, but also additional transparency, which is imperative for deployment in critical situations.

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8 Appendices

8.1 Appendix A.1: t-SNE Perplexity Fine-Tuning, KL Divergence

For the optimisation of the t-SNE embeddings on the MSAD dataset, it is imperative to explore how different perplexity values correspond with dimensionality reduction mappings. We measured the Kullback-Leibler divergence to evaluate

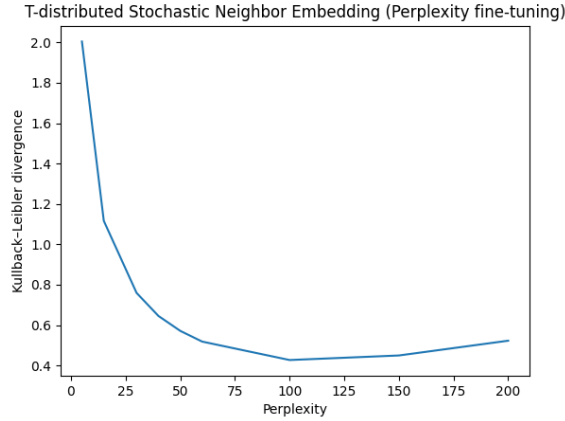


Fig. 17. Perplexity optimisation through Kullback-Leibler Divergence.

clustering quality, this is imperative as it enables us to show that our data is separable and thus may be used in model training. A lower KL divergence yields a greater representation of high-dimensional space in a 2-dimensional space. We can see that the optimal perplexity of 100 was achieved from Figure 17. The KL divergence is calculated with the following formula:

$$D_{KL}(P||Q) = \sum_{i,j} P_{ij} \log \frac{P_{ij}}{Q_{ij}}$$

Where:

P_{ij} represents the pairwise similarity between points in the high-dimensional space, and

Q_{ij} represents the corresponding similarity in the low-dimensional embedding. From the diagram of our results, it is apparent that a perplexity of 100 represents the minima and optimal result for our final embeddings.

8.2 Appendix A 2 - Training

Figure 18 represents the training run of the SlowFast binary classification model. The plot conveys that the model was able to train effectively, with improving generalisation until epoch 30. After epoch 30, the model shows signs of potential overfitting; future training could implement further transforms on the training data, regularization or expansion of the training dataset in order to improve generalisation. The figure represents the training run of the SlowFast multi-scenario classification model. The plot reveals rapid initial learning until the 10th epoch; it is at this point, further iterations yield repeated overfitting as generalisation ability declines. Akin to the binary case, further data, regularization or transforms could be deployed to enhance model learning abilities.

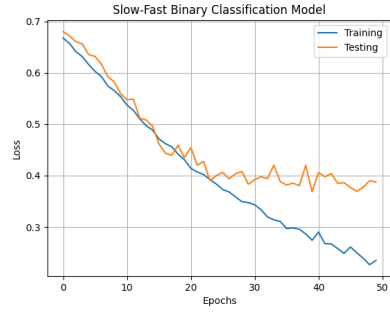
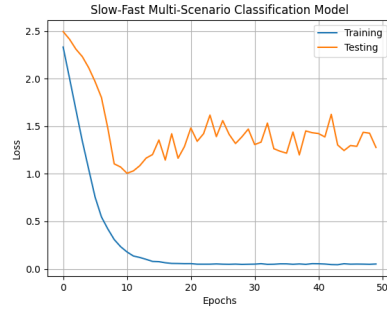
**Fig. 18.** Slow-Fast Binary.**Fig. 19.** Slow-Fast Multi-Scenario.**Fig. 20.** Training Runs for the Slow-Fast Architecture.

Figure 19 represents the training run of the SlowFast multi-scenario classification model. The plot reveals rapid initial learning until the 10th epoch, it is at this point, further iterations yield repaid overfitting as generalisational ability declines. Akin to the binary case, further data, regularisation or transforms could be deployed to enhance model learning abilities.

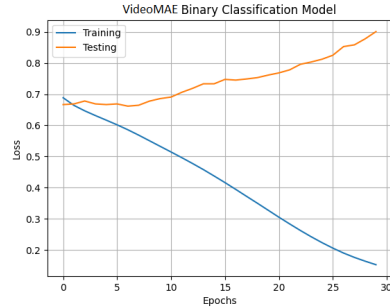
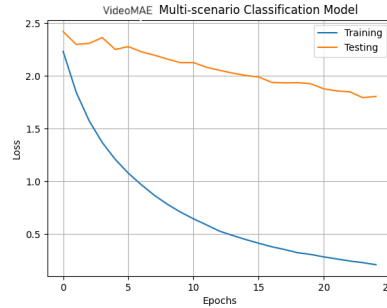
**Fig. 21.** VideoMAE Binary.**Fig. 22.** VideoMAE Multi-Scenario**Fig. 23.** Training Runs for the VideoMAE Transformer-Based Archetecture

Figure 21 represents the training run of the binary case for the VideoMAE transformer-based model. The model immediately overfits, this is caused by the model becoming increasingly specialised and learning features patterns that do not generalise well to unseen data.

Figure 22 represents the training run of the multi-scenario case of the VideoMAE transformer-based model. The plot shows greater training stability in contrast to the binary case. The model learns and generalises reasonably well, how-

ever the final loss is significantly higher than what would be desired for decent overall generalisation.

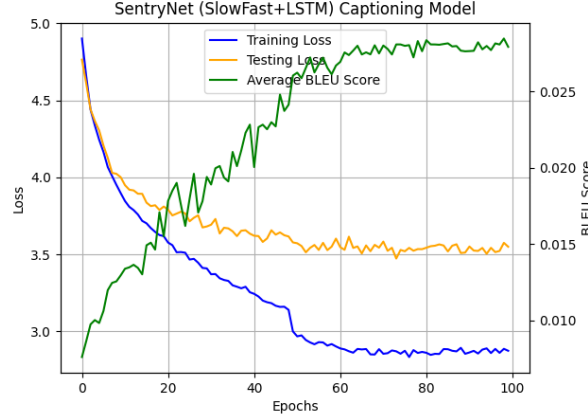


Fig. 24.

Figure 24 illustrates the training, testing loss alongside BLEU score of the SentryNet (SlowFast + LSTM) model. The graphic informs us that the model is successfully learning and improving BLEU score. However, in the early 60's the results begin to plateau, this suggests that the model has largely converged on the dataset.